A.C. Circuit Theory

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

A.C. Circuit Theory (A17698)

Short Title:	A.C. Circuit Theory		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electrical Engineering		
Author	PJCREGG		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module follows on from D.C. Circuit Theory. From RC & RL step response, square wave response is considered leading to sinusoidal response. Reactance and impedance are introduced and treated through phasor and complex representations. Frequency response of series RC, RL circuits is considered leading to series resonant RLC circuits. AC generators and motors, and transformers are treated with equivalent circuits. Teaching is carried out in the classroom and laboratory.

Programmes

	BEng (Hons) in Automation Engineering with Data Intelligence (`WD_EAUTO_B`)	2 / 3 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	2 / 3 / M
`ELTR-0015`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	2 / 3 / M
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	2 / 3 / M

Indicative Content

- Analysis: reactance, impedance, phasor, complex, resonance, power factor correction
- Electromagnetic Induction & AC machines, inductive energy, free wheeling diodes

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Treat an AC circuit and predict the amplitudes and phases of voltages and currents.
2. Produce magnitude and phase plots for series RC, RL & RLC circuits.
3. Describe the generation of sinusoidal AC voltage, based on a simple generator model.
4. Treat a transformer and AC motor through equivalent circuits.
5. Demonstrate a practical ability to develop, troubleshoot, test, analyse and report on AC circuits.

Learning and Teaching Methods

- Lecture / Lectorial
- Laboratory

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4
Continuous Assessment	30%	
In-Class Assessment	10%	1,2,3,4
Practical	20%	5

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Hughes, E. _Electrical and Electronic Technology_. 12th ed.. .: Pearson, 2016.

Supplementary Material

- Irwin, D.J. and R.M. Nelms. _Engineering Circuit Analysis_. 10th Ed.. New York: John Wiley, ..

Requested Resources

- ENGINEERING LAB: Electronics